Year 10 Foundation

# REVISION PACK

## Number 1

**ANSWERS**

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

## Number 1 · Working with fractions (Lesson 1.1)

| Q | Working | Answer |
|---|---|---|
| **adding fractions** | | |
| **1.1** | $\frac{12}{20} + \frac{5}{20} = \frac{17}{20}$ | $\frac{17}{20}$ |
| **1.2** | $\frac{20}{24} + \frac{3}{24} = \frac{23}{24}$ | $\frac{23}{24}$ |
| **1.3** | $\frac{21}{30} + \frac{5}{30} = \frac{26}{30}$<br>$= \frac{13}{15}$ | $\frac{13}{15}$ |
| **1.4** | $\frac{6}{21} + \frac{7}{21} = \frac{13}{21}$ | $\frac{13}{21}$ |
| **1.5** | $\frac{9}{18} + \frac{4}{18} = \frac{13}{18}$ | $\frac{13}{18}$ |
| **1.6** | $\frac{9}{24} + \frac{8}{24} = \frac{17}{24}$ | $\frac{17}{24}$ |
| **multiplying fractions** | | |
| **1.7** | $= \frac{15}{60}$<br>$= \frac{1}{4}$ | $\frac{1}{4}$ |
| **1.8** | $= \frac{28}{84}$<br>$= \frac{1}{3}$ | $\frac{1}{3}$ |
| **1.9** | $= \frac{30}{80}$<br>$= \frac{3}{8}$ | $\frac{3}{8}$ |
| **1.10** | $= \frac{6}{36}$<br>$= \frac{1}{6}$ | $\frac{1}{6}$ |

| | | |
|---|---|---|
| **1.11** | $= \frac{20}{120}$<br>$= \frac{1}{6}$ | $\frac{1}{6}$ |
| **1.12** | $= \frac{21}{126}$<br>$= \frac{1}{6}$ | $\frac{1}{6}$ |
| **dividing fractions** | | |
| **1.13** | $\frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$<br>$= 1\frac{7}{8}$ | $1\frac{7}{8}$ |
| **1.14** | $\frac{7}{8} \times \frac{3}{1} = \frac{21}{8}$<br>$= 2\frac{5}{8}$ | $2\frac{5}{8}$ |
| **1.15** | $\frac{9}{10} \times \frac{5}{3} = \frac{45}{30}$<br>$= \frac{3}{2}$<br>$= 1\frac{1}{2}$ | $1\frac{1}{2}$ |
| **1.16** | $\frac{5}{6} \times \frac{9}{4} = \frac{45}{24}$<br>$= \frac{15}{8}$<br>$= 1\frac{7}{8}$ | $1\frac{7}{8}$ |
| **1.17** | $\frac{2}{3} \times \frac{7}{4} = \frac{14}{12}$<br>$= \frac{7}{6}$<br>$= 1\frac{1}{6}$ | $1\frac{1}{6}$ |
| **1.18** | $\frac{8}{9} \times \frac{3}{2} = \frac{24}{18}$<br>$= \frac{4}{3}$<br>$= 1\frac{1}{3}$ | $1\frac{1}{3}$ |
| **fractions of amounts** | | |

| | | |
|---|---|---|
| **1.19** | $42 \div 6 = 7$<br>$7 \times 5 = 35$ | **35** |
| **1.20** | $81 \div 9 = 9$<br>$9 \times 2 = 18$ | **18** |
| **1.21** | $75 \div 5 = 15$<br>$15 \times 4 = 60$ | **60** |
| **1.22** | $56 \div 7 = 8$<br>$8 \times 3 = 24$ | **24** |
| **1.23** | $96 \div 8 = 12$<br>$12 \times 5 = 60$ | **60** |
| **1.24** | $57 \div 3 = 19$<br>$19 \times 2 = 38$ | **38** |
| **fractions, decimals and percentages** | | |
| **1.25** | **(a)** $2 \div 5 = 0.4$<br>**(b)** $0.4 \times 100 = 40$ | **(a) 0.4**<br>**(b) 40%** |
| **1.26** | **(a)** $7 \div 20 = 0.35$<br>**(b)** $0.35 \times 100 = 35$ | **(a) 0.35**<br>**(b) 35%** |
| **1.27** | **(a)** $5 \div 8 = 0.625$<br>**(b)** $0.625 \times 100 = 62.5$ | **(a) 0.625**<br>**(b) 62.5%** |
| **1.28** | **(a)** $3 \div 4 = 0.75$<br>**(b)** $0.75 \times 100 = 75$ | **(a) 0.75**<br>**(b) 75%** |
| **1.29** | **(a)** $9 \div 25 = 0.36$<br>**(b)** $0.36 \times 100 = 36$ | **(a) 0.36**<br>**(b) 36%** |
| **1.30** | **(a)** $1 \div 8 = 0.125$<br>**(b)** $0.125 \times 100 = 12.5$ | **(a) 0.125**<br>**(b) 12.5%** |

## Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

| Q | Working | Answer |
|---|---|---|
| **order of operations** | | |
| 2.1 | **(a)** $4 \times 5 = 20$<br>$7 + 20 = 27$<br>**(b)** $6^2 = 36$<br>$36 \div 4 = 9$ | **(a) 27**<br>**(b) 9** |
| 2.2 | **(a)** $3 \times 6 = 18$<br>$20 - 18 = 2$<br>**(b)** $7^2 = 49$<br>$49 - 9 = 40$ | **(a) 2**<br>**(b) 40** |
| 2.3 | **(a)** $18 \div 3 = 6$<br>$12 + 6 = 18$<br>**(b)** $8^2 = 64$<br>$64 \div 8 = 8$ | **(a) 18**<br>**(b) 8** |
| 2.4 | **(a)** $5 \times 3 = 15$<br>$6 + 15 = 21$<br>**(b)** $6^2 = 36$<br>$36 \div 9 = 4$ | **(a) 21**<br>**(b) 4** |
| 2.5 | **(a)** $4 \times 5 = 20$<br>$30 - 20 = 10$<br>**(b)** $7^2 = 49$<br>$49 - 10 = 39$ | **(a) 10**<br>**(b) 39** |
| 2.6 | **(a)** $24 \div 4 = 6$<br>$8 + 6 = 14$<br>**(b)** $8^2 = 64$<br>$64 \div 4 = 16$ | **(a) 14**<br>**(b) 16** |
| **rounding** | | |
| 2.7 | | **(a) 5.3**<br>**(b) 5.27** |

| | | |
|---|---|---|
| **2.8** | | **(a) 18.5**<br>**(b) 18.5** |
| **2.9** | | **(a) 9.1**<br>**(b) 9.08** |
| **2.10** | | **(a) 3.7**<br>**(b) 3.66** |
| **2.11** | | **(a) 27.4**<br>**(b) 27.4** |
| **2.12** | | **(a) 5.0**<br>**(b) 5.00** |